

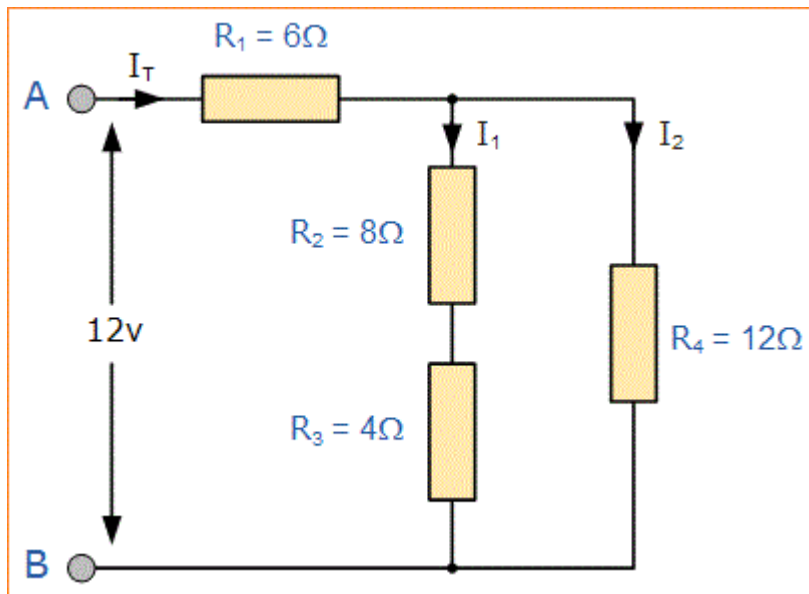
**LAPORAN PRATIKUM**  
**PRAKTIKUM ELEKTRONIKA DIGITAL**  
**(modul 4 Rangkaian Listrik)**

**Dosen Pengampu:**  
**NOVRINDAH ALVI HASANAH,M.Kom**

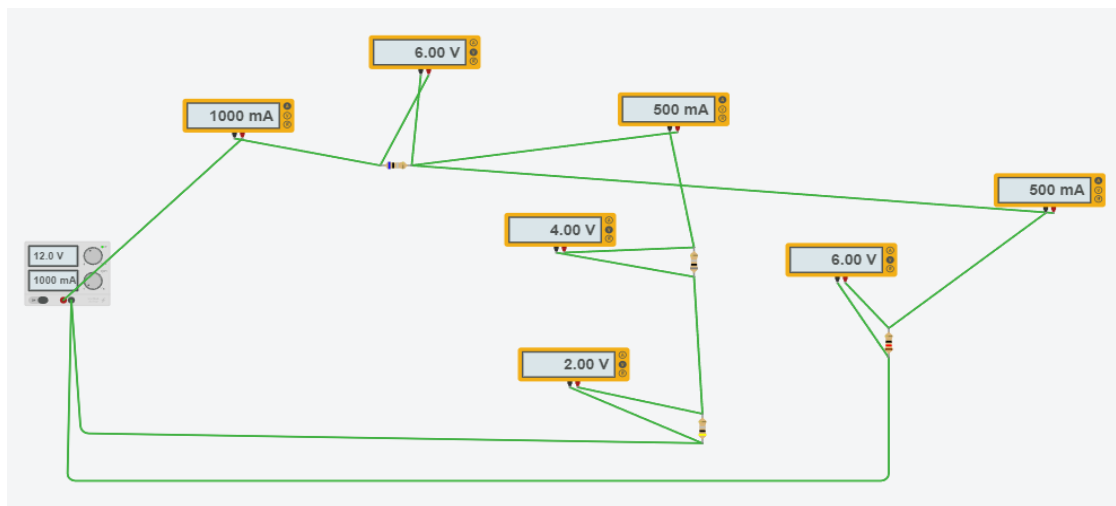


**Disusun Oleh:**  
**Rahmat Enomoto (230605110010)**

**PROGRAM STUDI**  
**TEKNIK INFORMATIKA**  
**FAKULTAS SAINS DAN TEKNOLOGI**  
**UNIVERSITAS ISLAM NEGERI MAULANA MAILK IBRAHIM MALANG**  
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Buat rangkaian seperti pada rangkaian berikut kemudian hitunglah  $I_T$  dan  $R_{total}$  menggunakan rumus yang telah dipelajari



| No. | Rangkaian | Hasil Pengukuran                                                                | Hasil Rumus                                                                     |
|-----|-----------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1.  | Tegangan  | $V_{R1} = 6V$<br>$V_{R2} = 4V$<br>$V_{R3} = 2V$<br>$V_{R4} = 6V$<br>$V_T = 12V$ | $V_{R1} = 6V$<br>$V_{R2} = 4V$<br>$V_{R3} = 2V$<br>$V_{R4} = 6V$<br>$V_T = 12V$ |
| 2.  | Arus      | $I_1 = 500mA$<br>$I_2 = 500mA$<br>$I_T = 1000mA$                                | $I_1 = 500mA$<br>$I_2 = 500mA$<br>$I_T = 1000mA$                                |

- Ukur arus dan tegangan pada masing-masing resistor dan kapasitor!  
Bandingkan dengan hasil rumus dan catat pada table

$$R_{\text{Seri } R_2 \text{ dan } R_3} = R_2 + R_3 = 8 + 4 = 12 \text{ ohm}$$

$$R_{\text{Paralel } R_2 R_3 + R_4} = \frac{1}{\frac{1}{12} + \frac{1}{12}} = \frac{2}{12} = \frac{1}{6} = 6 \text{ ohm}$$

$$R_{\text{total}} = R_1 + R_{\text{paralel}} = 6 + 6 = 12 \text{ ohm}$$

$$I_{\text{total}} = V/R = 12/12 = 1 \text{ A} = 1000 \text{ mA}$$

$$V_1 = I \times R_1 = 1 \times 6 = 6 \text{ V}$$

$$V_{R_2 R_3 R_4} = I \times R_{R_2 R_3 R_4} = 1 \times 6 = 6 \text{ V}$$

$$I_{R_4} = V_{R_2 R_3 R_4} / R_4 = 6/12 = 0,5 \text{ A} = 500 \text{ mA}$$

$$I_{R_2 R_3} = V_{R_2 R_3 R_4} / R_{R_2 R_3} = 6/12 = 0,5 \text{ A} = 500 \text{ mA}$$

$$V_{R_2} = I \times R_2 = 0,5 \times 8 = 4 \text{ V}$$

$$V_{R_3} = I \times R_3 = 0,5 \times 4 = 2 \text{ V}$$